

- P and Q are masses.

The distance between the centre of mass P and the centre of the right-hand eye is p , as shown in Fig. 1.2.

The distance between the centre of mass Q and the centre of the right-hand eye is q , as shown in Fig. 1.2.

The angle between the string and the rod is θ .

Use some of the adhesive putty to attach Q to the rod so that q is approximately 26 cm.

- Use some of the adhesive putty to attach P to the rod. Adjust the position of P and the position of the stand with the pulley so that the rod is parallel to the bench and θ is approximately 45° .

Do not move the stands for the remainder of the experiment.

- Measure and record θ , p and q .

$$\theta = \dots\dots\dots^\circ$$

$$p = \dots\dots\dots$$

$$q = \dots\dots\dots$$

[2]

- (ii) The moment of the force about the eye due to mass P is T_P .

The moment of the force about the eye due to mass Q is T_Q .

The values of T_P and T_Q are given by:

$$T_P = 5Wp \quad \text{and} \quad T_Q = 7Wq$$

where W has the value 0.0981 N.

Calculate T_P and T_Q .

$$T_P = \dots\dots\dots$$

$$T_Q = \dots\dots\dots$$

[1]

